

ZEPHYR TECHNOLOGIES AND SOLUTIONS PVT. LTD.

BUSINESS ANALYTICS INTERNSHIP REPORT

(03/06/2024 – 03/07/2024)



SUBMITTED BY,
SHRIDHAR SHETTY,
BUSINESS ANALYST INTERN,
ZEPHYR TECHNOLOGIES AND SOLUTIONS PVT. LTD.
MANGALORE.

SUBMITTED TO,
GOKULDEV K,
OPERATIONS MANAGER & DATA SCIENTIST
ZEPHYR TECHNOLOGIES AND SOLUTIONS PVT. LTD.
MANGALORE.

TABLE OF CONTENTS

Sl. No.	Content	Page no.
1	Executive Summary	4
2	Introduction	5
3	Description Of the Internship	5
4	Learning Outcomes (Projects)	10
5	Conclusion	30
6	References	30

ACKNOWLEDGEMENT

I am profoundly grateful to all those who made it possible for me to complete this internship.

Firstly, I am deeply indebted to Zephyr Technologies & Solutions Pvt. Ltd. for granting me the opportunity to undertake this internship and for providing invaluable support and resources throughout the period. I would like to extend special thanks to my supervisor, Mr. Gokuldev K, whose guidance and expertise were instrumental in the successful completion of this internship. His insightful feedback and unwavering support have significantly enriched my learning experience.

I would also like to express my sincere gratitude to the entire team at Zephyr Technologies & Solutions Pvt. Ltd., who warmly welcomed me and generously shared their knowledge and experience. Their willingness to impart insights and their collaborative spirit greatly enhanced my practical understanding of the industry. During my time at Zephyr Technologies & Solutions Pvt. Ltd., I had the privilege of working on challenging projects and collaborating with talented colleagues.

Additionally, I would like to acknowledge the steadfast support of my family and friends, whose encouragement and motivation kept me focused and determined throughout this journey.

Lastly, I extend my thanks to everyone who, directly or indirectly, contributed to this experience. Your support has been immensely valuable, and I am truly grateful.

Thank you all.

Shridhar Shetty

EXECUTIVE SUMMARY

This report provides a comprehensive overview of my internship as a Business Analyst at Zephyr Technologies & Solutions Pvt. Ltd., where I gained practical experience in various strategic and analytical frameworks, as well as data analysis tools. The internship was a valuable opportunity to apply theoretical knowledge to real-world business scenarios and enhance my professional skills.

During the internship, I learned to conduct SWOT (Strengths, Weaknesses, Opportunities, and Threats) analysis and evaluate it using TOWS (Threats, Opportunities, Weaknesses, and Strengths) metrics, providing a robust approach to strategic planning. I also performed PESTLE (Political, Economic, Social, Technological, Legal, and Environmental) analysis to assess the macro-environmental factors affecting the business landscape.

In addition, I conducted brand and market analyses to understand the competitive positioning and market dynamics of various products and services. My exposure to Lean Six Sigma methodologies, particularly the DMAIC (Define, Measure, Analyse, Improve, Control) process, and the Kaizen process, enabled me to identify and implement continuous improvement strategies. I also explored the differences between DMAIC and Kaizen processes, highlighting their unique applications and benefits.

Furthermore, I utilized Kano analysis to assess customer satisfaction by performing surveys and visualizing the results using Power BI. My proficiency in Power BI and Python for data analysis was significantly enhanced, allowing me to effectively analyse and interpret complex data sets.

Overall, this internship provided me with a solid foundation in strategic analysis and data-driven decision-making, equipping me with essential skills for my future career.

INTRODUCTION

Zephyr Technologies Mangalore is the Brainchild of two friends- Karun Lal and Samah, equipped with skills specialising in Branding, building websites, Graphic designs and Digital Media Contents we are set out to change the market of advertising and social media. Zephyr Technologies is a dynamic player in the technology solutions industry, specializing in providing innovative and tailored services to meet the evolving needs of businesses across various sectors. Established with a commitment to excellence and client satisfaction, Zephyr Technologies and Solutions Pvt. Ltd. Distinguishes itself through its comprehensive suite of offerings designed to optimize operational efficiencies, enhance cybersecurity measures and harness the power of advanced data analytics.

Why pick us? We do Brand Identity, Website design, Packaging and Marketing Communications Design, we don't just mean creating logos, templates and prototypes. We design your vision and principles. We take your thoughts and ideas and make it a measurable reality.

DESCRIPTION OF THE INTERNSHIP

Market Analysis

Market analysis is a comprehensive assessment of the market within a specific industry. It involves evaluating various factors that influence market dynamics, such as market size, customer demographics, competition, economic environment, and market trends. The primary objective of market analysis is to understand the current market conditions and predict future trends, helping businesses make informed decisions. Key components of market analysis include:

1. **Market Size and Growth:** Estimating the size of the market in terms of revenue and volume, and analysing historical growth patterns to predict future trends.
2. **Market Segmentation:** Dividing the market into distinct groups based on characteristics such as demographics, geographic location, and behavioural patterns.
3. **Target Market:** Identifying the specific group of consumers that the business aims to serve.
4. **Customer Analysis:** Understanding customer needs, preferences, and buying behaviour.
5. **Competitive Analysis:** Assessing the strengths and weaknesses of current and potential competitors.
6. **Economic Environment:** Evaluating economic factors such as inflation, employment rates, and economic growth that can impact the market.
7. **Regulatory Environment:** Understanding laws and regulations that govern the industry.
8. **Technological Trends:** Analysing technological advancements and innovations that could influence the market.

Brand Analysis

Brand analysis is the process of evaluating a brand's position in the market and its effectiveness in connecting with customers. This analysis helps businesses understand how their brand is perceived by their target audience and how it compares to competitors. Key components of brand analysis include:

1. **Brand Identity:** Assessing elements such as brand name, logo, tagline, and overall brand design to ensure they convey the desired message and values.
2. **Brand Positioning:** Understanding how the brand is positioned in the market relative to competitors, and whether it effectively communicates its unique value proposition.
3. **Brand Equity:** Measuring the brand's value, which includes brand awareness, brand loyalty, perceived quality, and brand associations.
4. **Customer Perception:** Gathering insights into how customers perceive the brand, including their emotional connection and overall satisfaction.
5. **Competitive Benchmarking:** Comparing the brand with key competitors to identify strengths and areas for improvement.
6. **Brand Performance:** Evaluating the brand's performance in terms of sales, market share, and profitability.
7. **Brand Communication:** Analysing the effectiveness of marketing and communication strategies in conveying the brand's message and engaging with the target audience.
8. **Customer Experience:** Understanding how customers interact with the brand across various touchpoints and the overall experience they have.

SWOT Analysis

SWOT Analysis is a strategic planning tool used to identify and evaluate the strengths, weaknesses, opportunities, and threats related to a business or project. It helps organizations understand their internal and external environments to make informed decisions.

- **Strengths:** Internal attributes and resources that support a successful outcome. Examples: strong brand reputation, skilled workforce, robust financial health.
- **Weaknesses:** Internal factors that might hinder success. Examples: limited resources, lack of expertise, poor location.
- **Opportunities:** External factors that the organization can capitalize on. Examples: market growth, technological advancements, favourable regulations.
- **Threats:** External factors that could cause trouble for the business. Examples: new competitors, economic downturns, changing consumer behaviour.

TOWS Metrics

TOWS Metrics is an extension of SWOT analysis, focusing on how the identified strengths, weaknesses, opportunities, and threats can be matched to develop strategic options. It involves creating strategies by matching external opportunities and threats with internal strengths and weaknesses. The TOWS matrix helps in the formulation of four types of strategies:

- **SO (Strengths-Opportunities) Strategies:** Use internal strengths to take advantage of external opportunities.

- **WO (Weaknesses-Opportunities) Strategies:** Improve weaknesses by taking advantage of external opportunities.
- **ST (Strengths-Threats) Strategies:** Use strengths to mitigate or avoid external threats.
- **WT (Weaknesses-Threats) Strategies:** Defensive strategies aimed at reducing internal weaknesses and avoiding external threats.

PESTLE Analysis

PESTLE Analysis is a tool used to analyse the macro-environmental factors that may impact an organization. It helps in understanding the external environment and the potential challenges and opportunities it presents.

- **Political Factors:** Government policies, regulations, trade restrictions, and political stability. Examples: tax policies, labour laws, environmental regulations.
- **Economic Factors:** Economic conditions and trends that influence business operations. Examples: inflation rates, interest rates, economic growth, unemployment rates.
- **Social Factors:** Societal and cultural aspects that affect business. Examples: demographic trends, cultural norms, health consciousness, lifestyle changes.
- **Technological Factors:** Technological advancements and innovations that impact the industry. Examples: automation, research and development, technological infrastructure.
- **Legal Factors:** Legal environment and regulations that affect business operations. Examples: employment laws, consumer protection laws, antitrust laws.
- **Environmental Factors:** Environmental and ecological aspects that can influence business. Examples: climate change, environmental regulations, sustainability initiatives.

Lean Six Sigma

Lean Six Sigma is a methodology that combines the principles of Lean and Six Sigma to improve efficiency and quality in business processes. Lean focuses on eliminating waste and optimizing workflows, while Six Sigma aims to reduce variability and defects through data-driven decision-making.

DMAIC Process

DMAIC is a structured problem-solving methodology within Six Sigma used to improve existing processes. It stands for Define, Measure, Analyse, Improve, and Control:

1. **Define:** Identify the problem, project goals, and customer (internal and external) requirements.
 - Tools: Project charter, SIPOC (Suppliers, Inputs, Process, Outputs, Customers) diagram.
2. **Measure:** Collect data and establish baseline metrics to understand the current process performance.
 - Tools: Data collection plan, measurement system analysis, process mapping.
3. **Analyse:** Identify the root causes of defects or inefficiencies by analysing the data collected.
 - Tools: Root cause analysis, Pareto charts, hypothesis testing.

4. **Improve:** Develop and implement solutions to address the root causes and improve process performance.
 - Tools: Brainstorming, design of experiments, pilot testing.
5. **Control:** Monitor the improved process to ensure that the improvements are sustained over time.
 - Tools: Control charts, process documentation, standard operating procedures (SOPs).

Kaizen Process

Kaizen is a Japanese term meaning "continuous improvement." It focuses on making small, incremental changes to improve efficiency and quality. The Kaizen process involves the following steps:

1. **Identify Opportunities:** Look for areas in the process where improvements can be made.
2. **Plan:** Develop a plan for implementing the improvements.
3. **Implement:** Execute the plan and make the necessary changes.
4. **Review:** Evaluate the results of the changes to ensure they have achieved the desired improvements.
5. **Standardize:** Document the new process and ensure that it becomes the standard way of working.
6. **Repeat:** Continuously look for new opportunities to improve.

Kaizen emphasizes a culture of continuous improvement where all employees are encouraged to contribute ideas for enhancing processes.

Kano Analysis

Kano Analysis is a model used to prioritize customer requirements based on their impact on customer satisfaction. It classifies features into different categories to help businesses understand which attributes will delight customers and which are simply expected.

1. **Basic Needs (Must-Be):** These are fundamental features that customers expect. If these needs are not met, customers will be dissatisfied. However, meeting these needs does not significantly increase satisfaction.
2. **Performance Needs (One-Dimensional):** These are the features that customer explicitly desire. Satisfaction increases proportionally with the level of performance or quality of these features.
3. **Excitement Needs (Delighters):** These are features that customers do not expect but are delighted when they are present. These features can significantly boost customer satisfaction.
4. **Indifferent Needs:** These are features that do not significantly impact customer satisfaction whether they are present or absent.
5. **Reverse Needs:** These are features that can lead to dissatisfaction if present in excess. Different customers may have varying opinions about these features.

Kano Analysis Process:

1. **Identify Customer Requirements:** Gather customer needs and expectations through surveys, interviews, or focus groups.
2. **Classify Features:** Categorize each feature based on customer feedback into one of the Kano categories.
3. **Prioritize Features:** Focus on improving or adding features that fall into the Performance and Excitement categories to maximize customer satisfaction.

Power BI

- **Objective:** Transform raw data into meaningful insights through data visualization.
- **Key Features:**
 - **Data Connectivity:** Connect to various data sources.
 - **Data Transformation:** Clean and shape data using Power Query.
 - **Data Visualization:** Create interactive charts, graphs, and maps.
 - **Dashboards and Reports:** Build and share interactive dashboards and reports.
 - **AI and Machine Learning:** Utilize built-in AI capabilities for data insights.
 - **Real-time Data:** Monitor and analyse streaming data.
 - **Mobile Accessibility:** Access data and reports on mobile devices.
- **Benefits:**
 - **Ease of Use:** Intuitive interface for users of all skill levels.
 - **Scalability:** Suitable for businesses of all sizes.
 - **Integration:** Seamless integration with other Microsoft products.
 - **Security:** Robust security features to protect data.

LEARNING OUTCOMES (Projects)

1. Market Analysis



Fire Bolt Brand

Presented by,
TEJAS B
Business Analyst Intern
Zephyr Technologies and Solutions Pvt Ltd

Content

- Introduction.
- Market Analysis.
- SWOT Analysis.
- PESTEL Analysis.

FIREBOLTT

Health
Assistant



Introduction to Fireboltt Smart Watch Company

Fireboltt is an innovative smart watch brand that is making waves in the wearable technology industry which was founded by Arnav Kishore in the year 2015. With a focus on cutting-edge design, advanced features, and unparalleled user experience, Fireboltt is poised to become a leading player in the competitive smart watch market. Fireboltt is committed to empowering its customers with the tools they need to stay connected, active, and in control of their digital lives.



Market Analysis



2. SWOT Analysis



Strengths

- | | | |
|--|---|---|
| <p>1 Innovative Design</p> <p>Firebolt's smart watches feature a sleek, modern design that sets them apart from the competition. With a focus on ergonomics and aesthetics, the company's products are both visually appealing and comfortable to wear.</p> | <p>2 Advanced Features</p> <p>Firebolt's smart watches are packed with a wide range of advanced features, including heart rate monitoring, GPS tracking, sleep tracking, and more. This comprehensive suite of capabilities allows users to better manage their health, fitness, and daily activities.</p> | <p>3 Robust Ecosystem</p> <p>Firebolt has built a robust ecosystem of mobile apps, cloud-based services, and integration with popular fitness and wellness platforms. This seamless integration across devices and platforms enhances the overall user experience and drives customer loyalty.</p> |
|--|---|---|

Weaknesses

Battery Life

Despite the advanced features, some of Firebolt's smart watches have been criticized for their relatively short battery life, which can be a concern for users who rely on their devices for extended periods of time.

Limited Availability

Firebolt's smart watches are currently only available in select markets, which can limit the company's reach and growth potential. Expanding distribution channels and global availability will be crucial for the brand's long-term success.

Steps tracking and integrating the device with different platforms

While Firebolt's smart watches offer premium features, the steps tracking needs improvement in the devices, so that we can get accurate data in terms of health and fitness. The devices must have the feature of integrating it with different platforms like google fit etc., which will be helpful to record all the data in the single platform.





Opportunities

Wearable Tech Boom

The global wearable technology market is experiencing rapid growth, driven by increasing consumer demand for connected devices that can seamlessly integrate with their digital lives. Firebolt is well-positioned to capitalize on this trend and expand its market share.

Health and Fitness Focus

With a growing emphasis on personal health and wellness, there is a significant opportunity for Firebolt to further differentiate its smart watches as premium fitness and wellness devices. Integrating more advanced health tracking features and partnering with fitness brands can help the company stand out in the crowded market.

Emerging Markets

Fire - Bolt aims to expand globally by entering UAE and Dubai markets. Untapped regions offer growth opportunities, and tailoring strategies to local preferences can drive sustainable expansion.



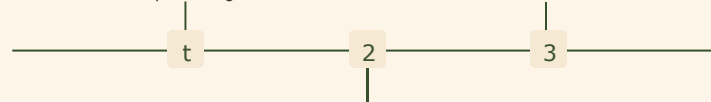
Threats

Intense Competition

Firebolt operates in a highly competitive smart watch market, with established tech giants and nimble startups vying for market share. The company must continuously innovate and stay ahead of the curve to differentiate its offerings and maintain a competitive edge.

Regulatory Challenges

As Firebolt expands its global footprint, it may face a complex web of regulatory and compliance requirements, particularly in areas related to data privacy, health and fitness tracking, and product safety standards. Navigating these challenges will be crucial for the company's long-term success.



Rapid Technological Change

The wearable technology industry is characterized by rapid advancements in hardware, software, and connectivity. Firebolt must remain agile and adaptable to ensure its products remain cutting-edge and relevant in the face of constantly evolving consumer preferences and technological trends.

3. Competitor Analysis



Competitors

- Apple - Renowned for the Apple Watch series, which is popular for its high-end design, advanced health and fitness tracking features, and seamless integration with the Apple ecosystem.
- Samsung - Known for the Galaxy Watch series, offering a variety of models with robust health and fitness tracking, long battery life, and integration with the Samsung ecosystem.
- Fitbit - Offers a wide range of fitness trackers and smartwatches known for their comprehensive health and fitness tracking capabilities, including heart rate monitoring, sleep tracking, and stress management features.
- Noise - Known for budget-friendly smartwatches and fitness bands, offering a variety of features including heart rate monitoring, sleep tracking, and long battery life, appealing to cost-conscious consumers.
- boAt - Another Indian brand, known for its range of affordable smartwatches and audio products, offering stylish designs and various health and fitness tracking features at competitive prices.

4. PESTLE Analysis

PESTEL Analysis

Political Factors:

- Regulations and standards.
- Trade Policies.

Economic Factors:

- Exchange Rates.
- Inflation Rates.

Social Factors:

- Brand Perception.
- Lifestyle Changes.

Technological Factors:

- Integration with other devices.
- Cybersecurity.

Environmental Factors:

- Resource Management.
- Sustainability.

Legal Factors:

- Data Privacy Laws.
- Consumer Protection Laws.

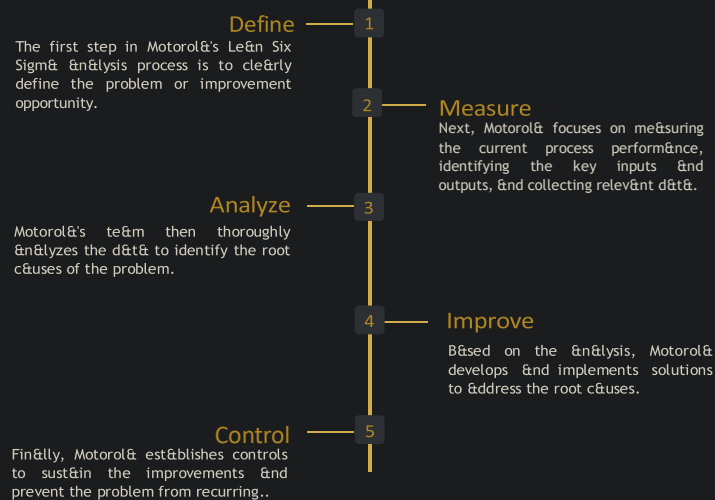
5. Innovative Strategies for Business Improvement: Lean Six Sigma and Kaizen

Motorola Company Overview

Motorola is a well-established technology company with a rich history dating back to 1928. Originally founded as a radio manufacturer, Motorola has evolved over the decades to become a global leader in the design and production of wireless communication devices, semiconductors, and other advanced electronic solutions. Motorola's diverse product portfolio includes smartphones, two-way radios, smart home devices, and critical infrastructure for public safety and commercial applications.

Headquartered in Chicago, Illinois, Motorola has a strong global presence with operations in multiple countries around the world. The company is known for its innovative approach to technology, continuously pushing the boundaries of what's possible and delivering cutting-edge solutions to its customers. Motorola's commitment to R&D and its focus on quality have been instrumental in maintaining its position as a trusted brand in the industry.

Lean Six Sigma Analysis Process for Motorola



Toyota Company Overview

Toyota is a renowned automotive company with a rich history dating back to 1937. Originally founded as a division of Toyoda Automatic Loom Works, Toyota has evolved over the decades to become a global leader in the design and production of automobiles, including cars, trucks, and hybrid vehicles. Toyota's diverse product portfolio includes the popular Camry, Corolla, Prius, and the luxury Lexus brand.

Headquartered in Toyota City, Japan, Toyota has a strong global presence with operations in numerous countries worldwide. The company is known for its innovative approach to manufacturing, pioneering practices like the Toyota Production System (TPS) and Lean manufacturing, which emphasize efficiency, quality, and continuous improvement. Toyota's commitment to sustainability and cutting-edge technology, such as hybrid and hydrogen fuel cell vehicles, has been instrumental in maintaining its position as a trusted brand in the industry. Through its focus on research and development, Toyota continues to push the boundaries of automotive innovation, delivering reliable and advanced solutions to its customers.



Kaizen Process at Toyota Company

1

Continuous Improvement

At the heart of the Toyota Production System (TPS) lies the Kaizen philosophy, which focuses on continuous, incremental improvements and empowers all employees to implement changes.

2

Employee Engagement

Toyota promotes Kaizen participation across all levels, fostering ownership, accountability, innovation, and problem-solving.

3

Data-Driven Decisions

Toyota's Kaizen uses a data-driven approach, ensuring improvements that are based on facts, leading to effective and sustainable outcomes.

Comparison of Lean Six Sigma and Kaizen

Lean Six Sigma

Lean Six Sigma is a data-driven methodology that focuses on reducing waste and variability to improve efficiency and quality. It emphasizes statistical analysis, process mapping, and project management to identify and eliminate defects.

Key Differences

While both Lean Six Sigma and Kaizen aim to improve efficiency and quality, Lean Six Sigma takes a more data-driven and project-based approach, while Kaizen emphasizes a cultural shift towards continuous improvement through employee engagement and empowerment.

Kaizen

Kaizen is a Japanese philosophy of continuous improvement that involves everyone in the organization, from top management to frontline employees. It emphasizes small, incremental changes to enhance productivity and reduce waste.

Complementary Approaches

Many organizations have found success by, using the data-driven analysis of Lean Six Sigma to identify areas for improvement and the continuous improvement mindset of Kaizen to sustain those improvements over time.

Benefits of Lean Six Sigma and Kaizen

1

Increased Efficiency

Lean Six Sigma and Kaizen methodologies focus on streamlining processes, reducing waste, and improving productivity, leading to enhanced operational efficiency and cost savings for organizations.

2

Quality Improvements

By identifying and eliminating defects, these approaches help organizations deliver higher-quality products and services, meet customer expectations, and improve overall customer satisfaction.

3

Employee Engagement

Lean Six Sigma and Kaizen encourage a culture of continuous improvement, where employees are empowered to identify and implement process improvements, fostering a sense of ownership and engagement.

4

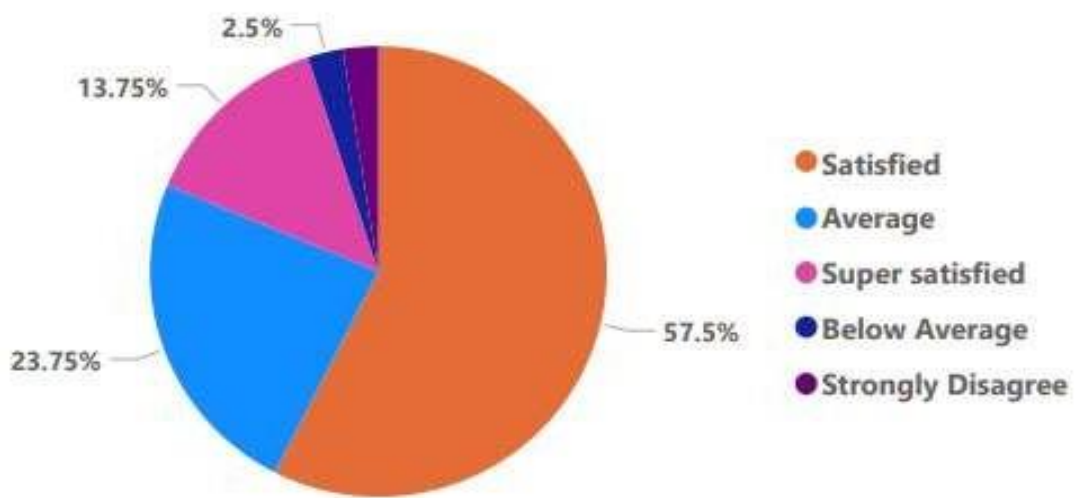
Competitive Advantage

The systematic and data-driven approach of these methodologies enables organizations to stay ahead of the competition, adapt to changing market conditions, and maintain a sustainable competitive edge.

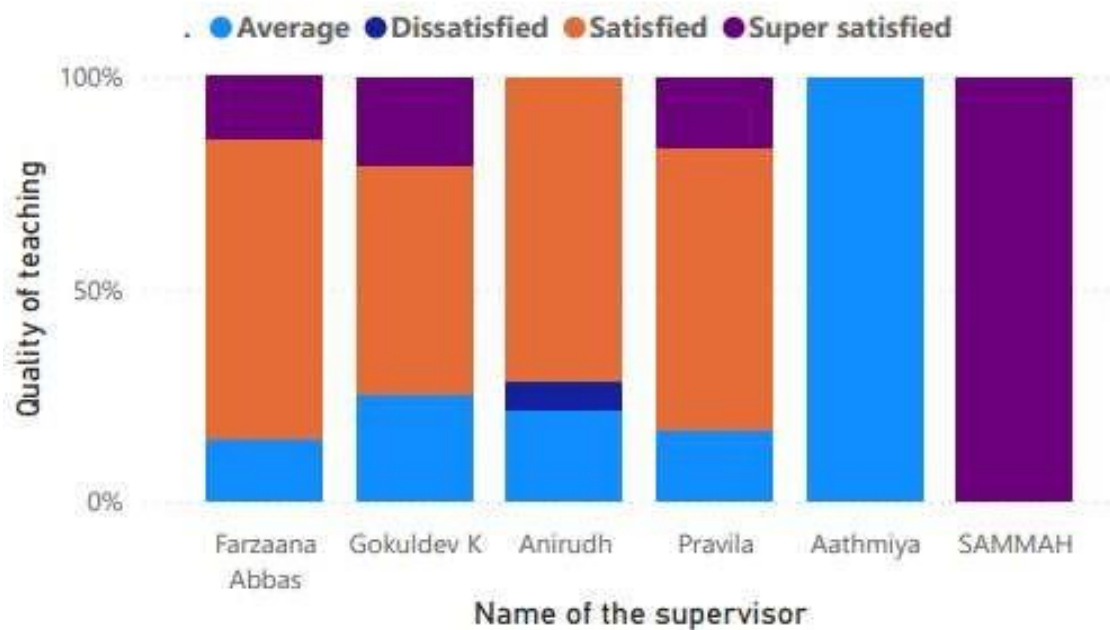
6. KANO ANALYSIS ON ZEPHYR TECHNOLOGIES AND SOLUTIONS PVT. LTD. USING POWER BI

Survey was conducted by our team of Business Analyst Interns, where we had circulated the google forms with the students and interns at Zephyr Technologies and Solutions Pvt. Ltd. To understand their experience at Zephyr Technologies and summarize them.

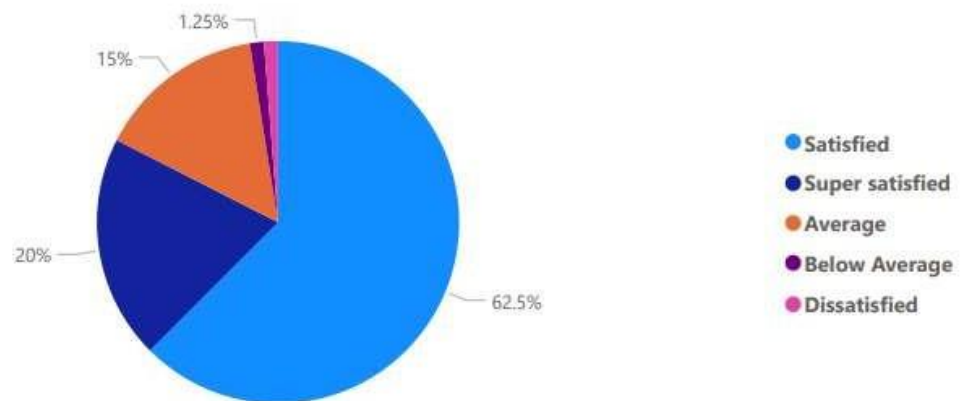
Are classroom facilities conducive to effective learning ?



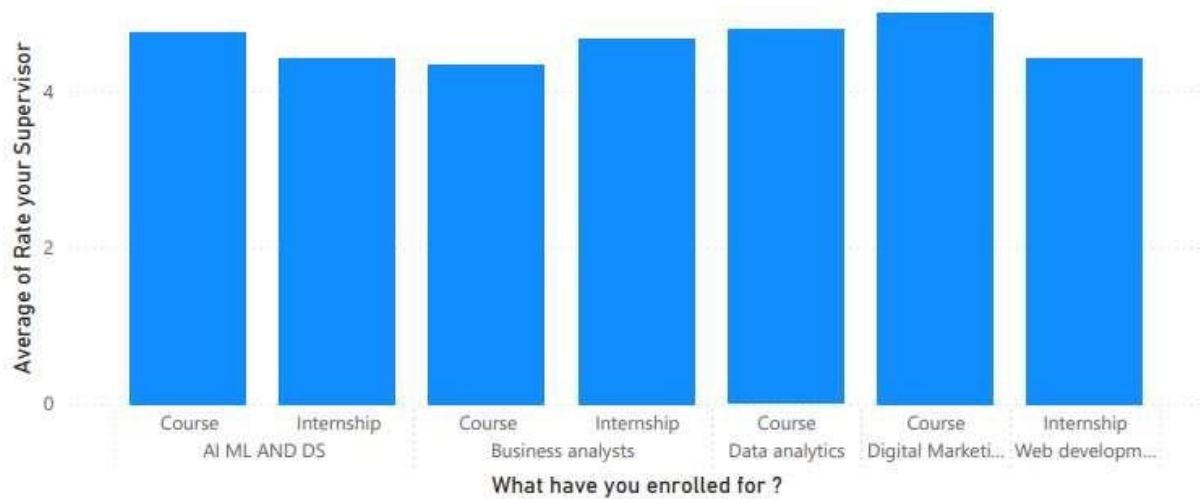
Quality of teaching by Name of the supervisor and .



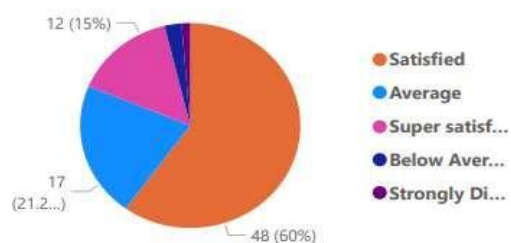
Do you feel there's a good support / transition program for students ?



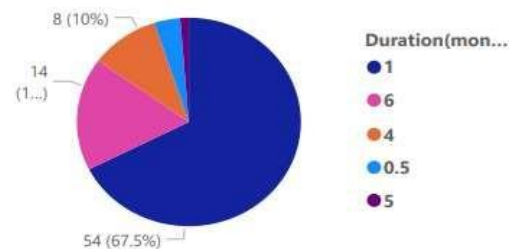
Rate your Supervisor by Name of the Course / Internship and What have you enrolled for ?



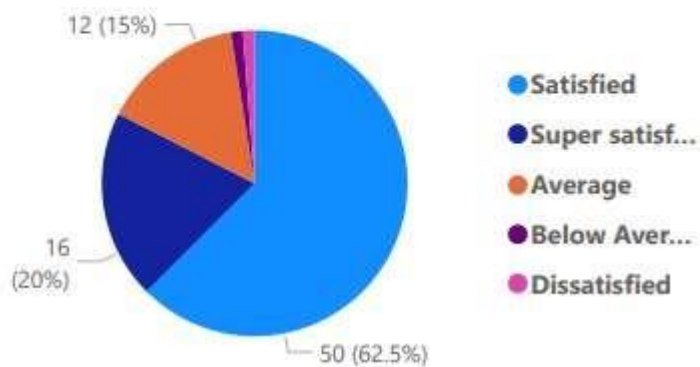
Do you find the course materials relevant and up-to-date ?



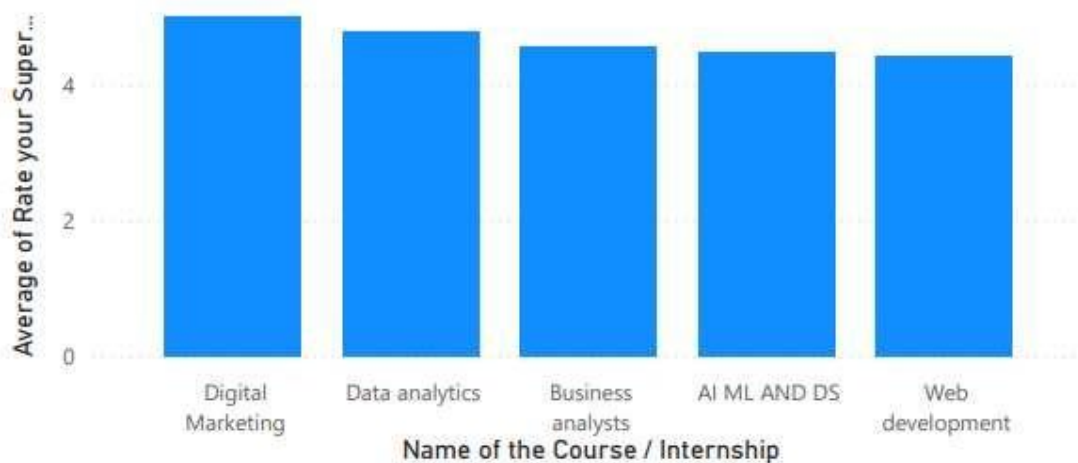
Duration of Course / Internship



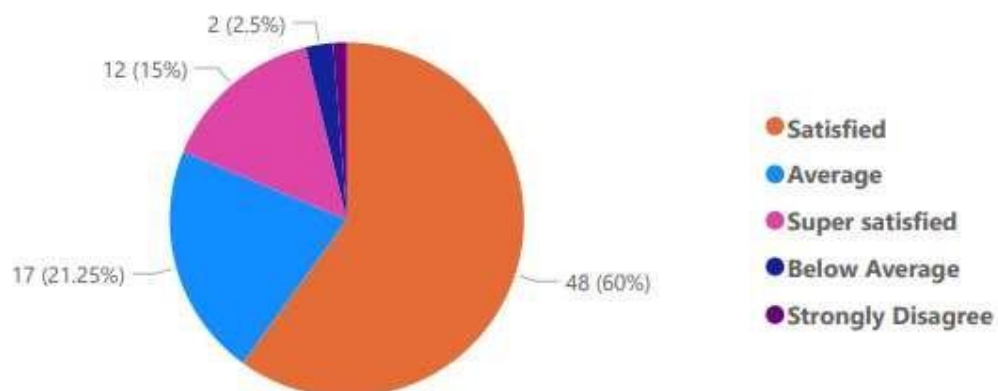
Do you feel there's a good support / transition program for students ?



Rate your Supervisor by Name of the Course / Internship

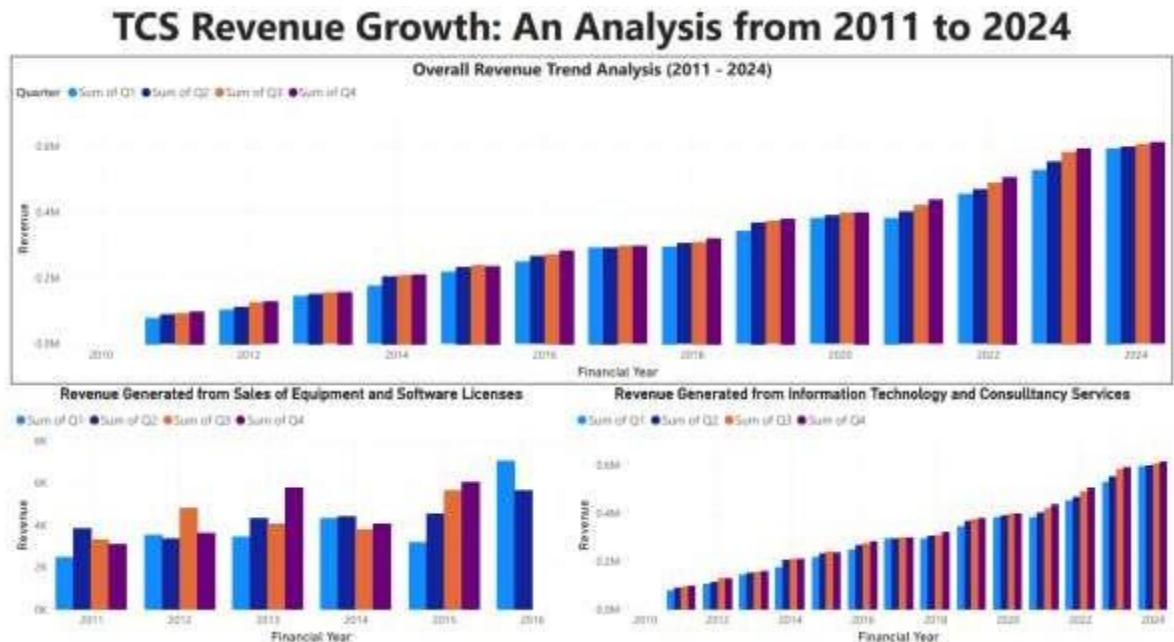


Do you find the course materials relevant and up-to-date ?

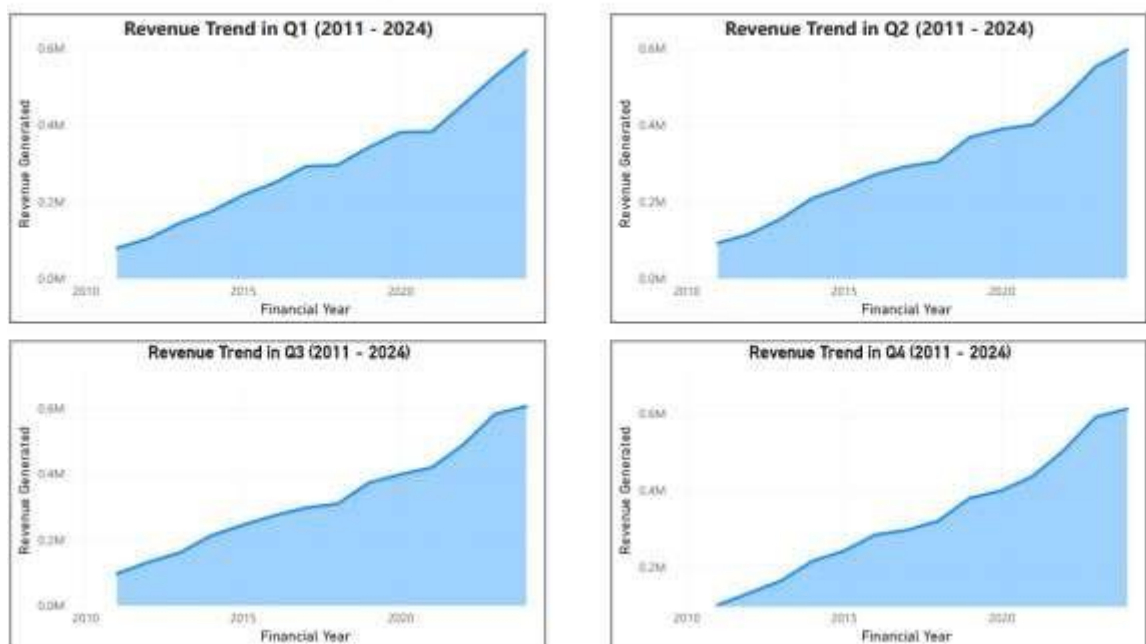


7. Kano Analysis on sales data of TCS company using Power BI and Python programming

A. Power BI



TCS Quarterly Revenue Trends (2011-2024)



B. Python Programming

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
```

```
data = pd.ExcelFile('/content/TC5data.xlsx')
```

```
for sheet_name in data.sheet_names:
    df = pd.read_excel('/content/TC5data.xlsx', sheet_name=sheet_name)
    csv_file_path = f'{sheet_name}.csv'
    df.to_csv(csv_file_path, index=False)
    print(f'Sheet {sheet_name} has been saved as {csv_file_path}')
```

```
Sheet Total Revenue has been saved as Total Revenue.csv
Sheet Quarterwise Total Revenue has been saved as Quarterwise Total Revenue.csv
Sheet IT and CS Revenue has been saved as IT and CS Revenue.csv
Sheet Equip. and SWL Revenue has been saved as Equip. and SWL Revenue.csv
Sheet Cost of Revenues has been saved as Cost of Revenues.csv
```

```
data = pd.read_csv('Total Revenue.csv')
data.head()
```

	Financial Year	Information Technology and Consultancy Services	Sale of Equipment and Software Licenses	Total Revenue
0	1Q11	79670.6	2502.2	82172.8
1	2Q11	88997.8	3866.1	92863.9
2	3Q11	93314.7	3318.8	96633.5
3	4Q11	98480.4	3094.4	101574.8
4	FY11	360463.5	12781.6	373245.1

Next steps: [Generate code with data](#) [View recommended plots](#)

```
data.isnull().sum()
```

```
Financial Year      0
Information Technology and Consultancy Services    0
Sale of Equipment and Software Licenses          43
Total Revenue      0
dtype: int64
```

```
data['Sale of Equipment and Software Licenses'].dropna(inplace=True)
```

```
data.isnull().sum()
```

```
Financial Year      0
Information Technology and Consultancy Services    0
Sale of Equipment and Software Licenses          43
Total Revenue      0
dtype: int64
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 70 entries, 0 to 69
Data columns (total 4 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Financial Year                       70 non-null    object
 1   Information Technology and Consultancy Services  70 non-null    float64
 2   Sale of Equipment and Software Licenses  27 non-null    float64
 3   Total Revenue                       70 non-null    float64
dtypes: float64(3), object(1)
memory usage: 2.3+ KB
```

```
data.describe().T
```

	count	mean	std	min	25%	50%
Information Technology and Consultancy Services	70.0	512070.217143	494594.211696	79670.6	234231.625	370960.00


```
data1 = pd.read_csv('/content/Quarterwise Total Revenue.csv')
data1.head()
```

	FY	Q1	Q2	Q3	Q4	Total Revenue
0	2011	82172.8	92863.9	96633.5	101574.8	373245.0
1	2012	107970.2	116334.9	132040.0	132593.2	488938.3
2	2013	148687.1	156207.6	160699.1	164301.0	629894.8
3	2014	179870.7	208772.4	212939.6	215510.9	818093.6
4	2015	221110.3	238164.8	245011.4	242198.0	946484.5

Next steps: [Generate code with data1](#) [View recommended plots](#)

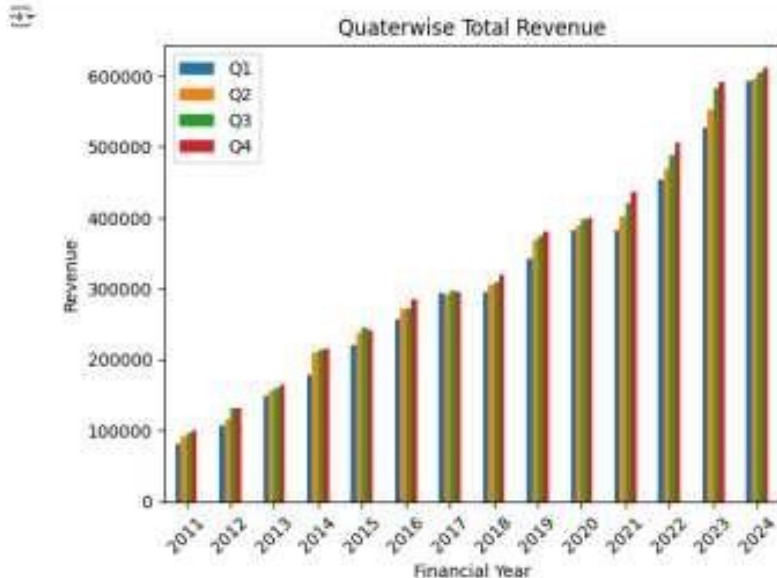
```
data1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14 entries, 0 to 13
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype  
---  --
 0   FY              14 non-null    int64  
 1   Q1              14 non-null    float64
 2   Q2              14 non-null    float64
 3   Q3              14 non-null    float64
 4   Q4              14 non-null    float64
 5   Total Revenue   14 non-null    float64
dtypes: float64(5), int64(1)
memory usage: 888.8 bytes
```

```
data1.describe().T
```

	count	mean	std	min	25%	50%	75%
FY	14.0	2.017500e+03	4.183300	2011.0	2014.250	2017.5	2020.7
Q1	14.0	3.048880e+05	153614.953565	82172.8	190180.600	294445.0	382845.0
Q2	14.0	3.186849e+05	154165.032463	92863.9	216870.500	299125.0	398455.0
Q3	14.0	3.283138e+05	158733.301089	96633.5	220957.550	303195.0	414747.5
Q4	14.0	3.345860e+05	161799.059371	101574.8	222162.675	308585.0	427652.5
Total	14.0	3.062445e+06	877650.436015	373245.0	1.000000e+06	1.000000e+06	1.000000e+06

```
data1.iloc[:,1:-1].plot(kind='bar')
plt.xticks(ticks=range(len(data1)), labels=range(2011, 2025), rotation=45)
plt.xlabel("Financial Year")
plt.ylabel("Revenue")
plt.title("Quarterwise Total Revenue")
plt.show()
```



```

FY = data1.iloc[:,0]
Q1 = data1.iloc[:,1]
Q2 = data1.iloc[:,2]
Q3 = data1.iloc[:,3]
Q4 = data1.iloc[:,4]

fig, axs = plt.subplots(2, 2, figsize=(10, 8))

axs[0, 0].plot(FY, Q1, 'r')
axs[0, 0].set_title('Revenue generated in Quarter 1')

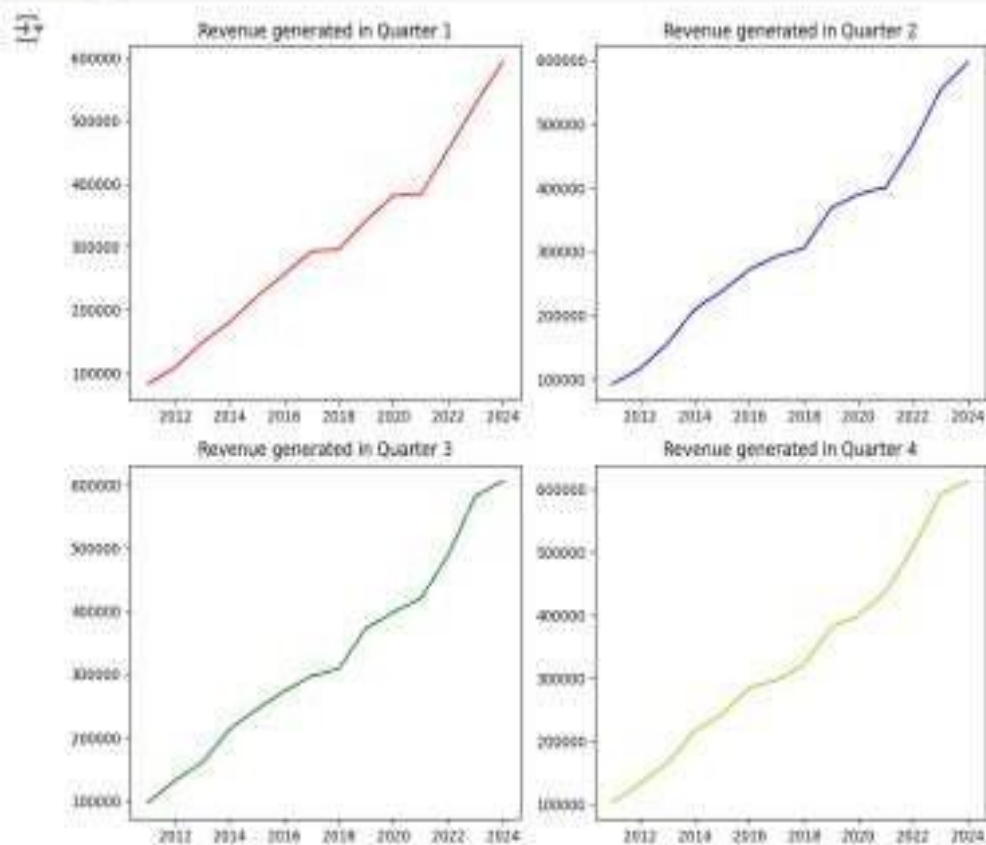
axs[0, 1].plot(FY, Q2, 'b')
axs[0, 1].set_title('Revenue generated in Quarter 2')

axs[1, 0].plot(FY, Q3, 'g')
axs[1, 0].set_title('Revenue generated in Quarter 3')

axs[1, 1].plot(FY, Q4, 'y')
axs[1, 1].set_title('Revenue generated in Quarter 4')

plt.tight_layout()
plt.show()

```




```
data2 = pd.read_csv('/content/IT and CS Revenue.csv')
data2.head()
```

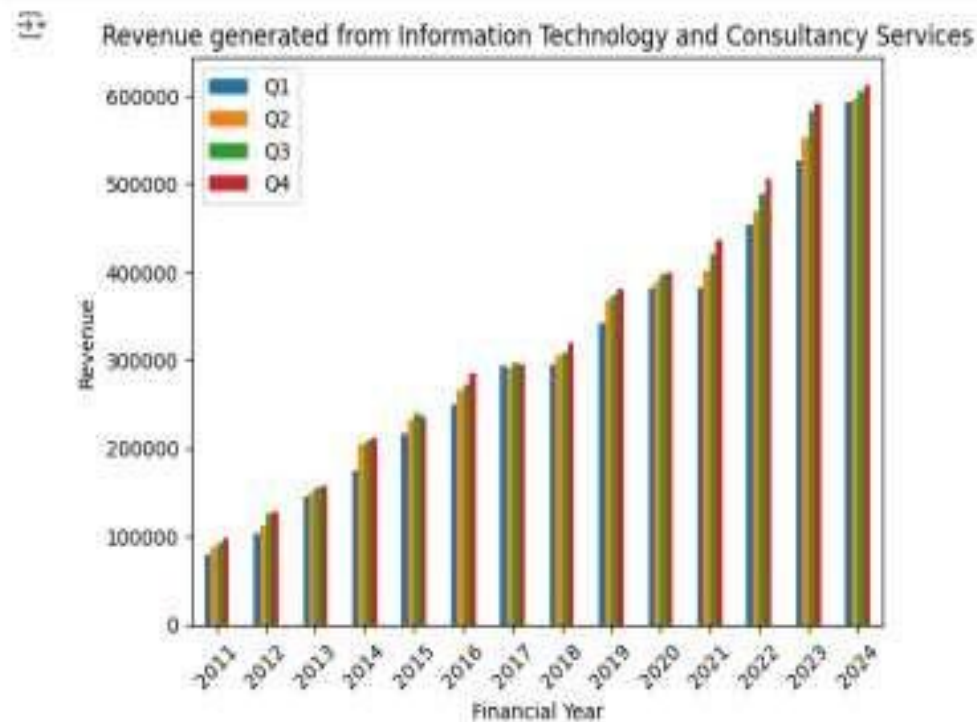
	FY	Q1	Q2	Q3	Q4	Total Revenue
0	2011	79670.6	88997.8	93314.7	98480.4	360463.5
1	2012	104442.1	112962.9	127230.3	128969.8	473605.1
2	2013	145213.7	151880.5	156630.0	158505.7	612229.9
3	2014	175517.8	205332.3	209124.6	211423.7	801398.4
4	2015	217928.4	233586.5	239355.1	236137.0	927017.0

Next steps:

[Generate code with data2](#)

[View recommended plots](#)

```
data2.iloc[:,1:-1].plot(kind='bar')
plt.xticks(ticks=range(len(data2)), labels=range(2011, 2025), rotation=45)
plt.xlabel("Financial Year")
plt.ylabel("Revenue")
plt.title("Revenue generated from Information Technology and Consultancy Services")
plt.show()
```



```
data3 = pd.read_csv('/content/Equip. and S&L Revenue.csv')
data3
```

	FY	Q1	Q2	Q3	Q4	Total Revenue
0	2011	2502.2	3866.1	3318.8	3094.4	12781.5
1	2012	3528.1	3372.0	4809.7	3623.4	15333.2
2	2013	3473.4	4327.1	4069.1	5795.3	17664.9
3	2014	4352.9	4440.1	3815.0	4087.2	16695.2
4	2015	3181.9	4568.3	5656.3	6061.0	19467.5
5	2016	7070.0	5646.8	0.0	0.0	12716.8

Next steps: [Generate code with data3](#) [View recommended plots](#)

```
data3.iloc[5,3:5] = 0
```

```
# change the datatype of Q3 and Q4 to float
data3['Q3'] = data3['Q3'].astype(float)
data3['Q4'] = data3['Q4'].astype(float)
```

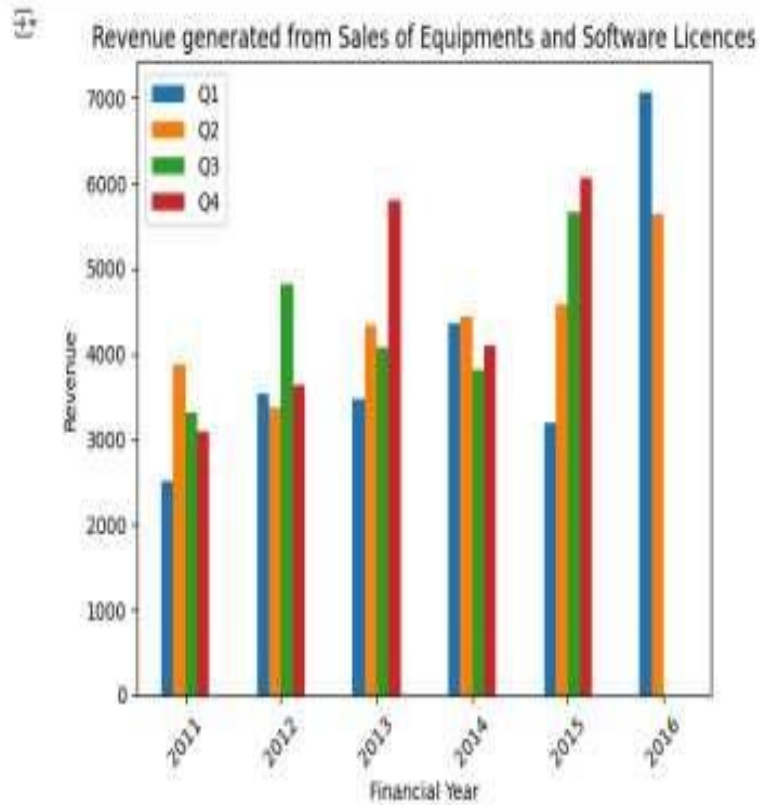
```
data3.isnull().sum()
```

```
FY      0
Q1      0
Q2      0
Q3      0
Q4      0
Total Revenue  0
dtype: int64
```

```
data3.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   FY          6 non-null     int64
1   Q1          6 non-null     float64
2   Q2          6 non-null     float64
3   Q3          6 non-null     float64
4   Q4          6 non-null     float64
```

```
data3.iloc[:,1:-1].plot(kind='bar')
plt.xticks(ticks=range(len(data3)), labels=range(2011, 2017), rotation=45)
plt.xlabel("Financial Year")
plt.ylabel("Revenue")
plt.title("Revenue generated from Sales of Equipments and Software Licences")
plt.show()
```



CONCLUSION

At the end of my internship, I am equipped with new skills that have enhanced my knowledge and provided valuable experience, which will be instrumental in my career. Specifically, I have learned how to conduct SWOT analysis and evaluate it using TOWS metrics, perform PESTLE analysis, carry out brand and market analysis, and apply Lean Six Sigma methodologies, including the DMAIC process. I have also gained proficiency in the Kaizen process, conducted Kano analysis through surveys, and visualized data using Power BI. Additionally, I have developed skills in data analysis with Power BI and Python.